

Taylor M. Giles

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Summary

Behavioral and data science expert with a PhD in Human Services Psychology and more than 6 years of experience designing and executing large-scale research, impact evaluations, and equity-centered data science. Skilled in both qualitative and quantitative methodologies, predictive analytics, and stakeholder engagement across public and private sectors. Adept at translating complex datasets into actionable insights that influence policy, product, and equity-centered strategy.

Core Competencies & Technical Skills

Quantitative & Qualitative Research | Predictive Modeling & Regression Analysis | Data Visualization (Tableau, PowerBI, Python, Matplotlib) | Equity & Disparity Evaluation | Mixed-Methods Research | Data Automation & Pipeline Development | Python, R, SQL, SPSS, STATA, NVivo | Stakeholder Engagement | Legislative Research & Policy Translation | Experimental Design | IRB Compliance & Ethical Data Use

Work Experience

Science Analyst

September 2022 - July 2025

Lexical Intelligence, LLC

Data Science & Automation:

- Designed and automated Python pipelines to extract, clean, and analyze over 20,000 lines of scientific publication data across more than 20 years.
- Developed regression models and predictive analytics using Python to detect emerging trends, providing evidence to guide strategic research planning and portfolio prioritization
- Built data dashboards and visualizations (heatmaps, bubble charts) using Pandas and Matplotlib for dissemination insights.

Equity Impact & Network Analysis:

- Combined Python and Cytoscape to map research impact and citation networks, highlighting gaps and opportunities in representation across research focus areas.
- Developed automated workflows for keyword extraction that improved reproducibility, equity lens analysis, and efficiency.

Stakeholder Engagement & Strategic Alignment:

- Partnered with senior analysts and data scientists to align modeling approaches with organizational research goals.
- Delivered executive-facing dashboards and reports translating complex technical outputs into strategic insights.

Science Analyst

July 2019 - September 2022

National Science Foundation (NSF)

Research Evaluation & Policy Analysis:

- Conducted statistical and qualitative analyses on STEM education programs using SPSS, Stata, and NVivo.
- Evaluated program efficacy and synthesized findings into high-impact reports and presentations for NSF leadership that guided strategic funding decisions and resource allocations across multi-million-dollar federal investments

Cross-Functional Collaboration & Data Storytelling:

- Designed Tableau dashboards to communicate performance insights to technical and non-technical stakeholders.
- Partnered with directors, reviewers, and grantees to ensure scientific rigor and alignment with agency metrics, driving evidence-based decision-making.

Legislative & Equity-Focused Research:

- Supported division-wide equity evaluation initiatives, improving decision-making around underrepresented populations in STEM.
- Analyzed policy impact and legislative language to assess systemic bias and improve programmatic equity

tracking.

Graduate Research Assistant

June 2016 - June 2018

University of Maryland, Baltimore County

Health Equity & Disparity Research:

- Led independent research examining the intersection of racial discrimination, poverty, and cardiovascular health outcomes.
- Analyzed data using SPSS and Stata to model health disparities in vulnerable populations.

Policy Impact & Stakeholder Engagement:

- Produced legislative-focused dissertation analyzing how language in health disparity bills affected progression through Congress.
- Delivered presentations to diverse stakeholders on findings related to structural racism and public health inequities.

Qualitative Research & Leadership:

- Led focus groups with high school students to assess school climate and voice, analyzing data with NVivo and Dedoose.
- Supervised and mentored undergraduates in research design, data analysis, and ethical research practices.

Evaluation Intern

October 2018 - July 2019

National Science Foundation (NSF)

- Conducted program evaluations using mixed methods to assess STEM education outcomes.
- Designed surveys and reported insights to NSF leadership to improve program processes and data reliability.

Research-to-Policy Collaboration Intern

January 2017 - July 2019

National Prevention Science Coalition

- Synthesized research into policy briefs and op-eds for legislative audiences.
- Supported rapid-response advocacy network linking researchers to federal lawmakers.

Policy Intern

June 2018 - August 2018

American Psychological Association (APA)

- Researched legislative policies on health equity and mental health access.
- Drafted letters and public comments on behalf of APA for key public policy proposals.

Education & Certifications

Ph.D., Human Services Psychology (Community) – University of Maryland, Baltimore County (2022)

M.A., Human Services Psychology (Clinical) – University of Maryland, Baltimore County (2019)

M.A., Counseling Psychology – Towson University (2016)

B.A., Psychology & Spanish (Cum Laude) – University of Dayton (2013)